Exercise - Week 12: MATH1061/7861

Duration: 30 minutes

2 Questions - 10 marks

Topics: Counting Selections, Probability, Binomial Coefficient

Student Name:

Student number: |__|__|__|__|__|__|__|__|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

Question:

(1) There are 5 females and 4 males on a list. Our goal is to choose one team of 3 people from this list.

(a) **(1 mark)** How many different teams can be chosen from the list?
Give your answer as an integer.

(b) **(2 marks)** If there is to be at least one person of each gender (male or female) on the team, how many different teams can be chosen from the list? Give your answer as an integer.

(c) **(2 marks)** What is the probability that a team chosen at random contains both a male and a female?
Express your answer as a rational number $\frac{a}{b}$ with $\gcd(a, b) = 1$.

(d) **(2 marks)** What is the probability that a team chosen at random contains only males?
Express your answer as a rational number $\frac{a}{b}$ with $\gcd(a, b) = 1$.

(2) **(3 mark)** What is the coefficient of x^3 in the expansion of $(2x - 3)^6$?